PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

Elevator Maintenance and Asset Tracking App

AN INTERNSHIP REPORT

Submitted by

Ann Olivia M - 20221COM0208

Under the guidance of,

Dr. Mohamed Shakir

BACHELOR OF TECHNOLOGY

IN

COMPUTER ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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APRIL 2026

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BONAFIDE CERTIFICATE

Certified that this report **"Elevator Maintenance and Asset Tracking App"** is a Bonafide work of **Ann Olivia M(20221COM0208),** who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER ENGINEERING (Artificial Intelligence and Machine Learning) during 2025-2026.

23.04.2026

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DECLARATION

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PLACE: BENGALURU

DATE: 23 - April 2026

INTERNSHIP COMPLETION CERTIFICATE

Date: 30/12/2025

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Presidency School of CSE,
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Through,
The Head of Department, Program Internship Coordinator, Supervisor and Internship Coordinator

Program: B.Tech in Computer Engineering (Artificial Intelligence and Machine Learning)
Presidency School of CSE,
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Subject: Requesting Permission for 8th semester Internship – reg.

Respected Sir/Madam,

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Thanking you.

Company's Details

Name: Axxonet Solutions India Private Limited

CIN (Corporate Identification Number) or MSME (Udayam Registration Number) :

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ACKNOWLEDGEMENTS

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Ann Olivia M

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Methodology, For the current project, the adopted methodology entails the creation of an entire stack web application, which makes use of advanced software engineering principles. The frontend of the application will be designed using ReactJS together with Material UI to enable interactive design of user interface elements. The backend takes care of data handling including processing and storing data in addition to data exchange using API calls. The design of the web application follows a modular approach where functions within the web application are implemented as independent modules including authentication, assets, maintenance, and requests.

One of the distinctive features of this system is the capacity to provide real-time information regarding the condition of the assets and maintenance activities being carried out. This system will assist users in understanding who the asset is, service history, planned maintenance activities according to the AMC strategy, and timing of services. Further, users will be able to generate service tickets (tokens), track service requests status, and schedule maintenance services when necessary. Furthermore, the inclusion of QR code functionality will ensure easy access to asset details and generating of service requests.

The testing process adopted in analyzing the functionality of the application involved functional testing, validation testing, and usability testing. Some of the scenarios tested included functional validity with both valid and invalid data, navigation, and testing of API response. The test results reveal that the application works perfectly well, with most of the functions doing their job. Nonetheless, there was a problem in validating the interface of the user and some delays in the response; however, these issues have been sorted out.

The outcomes from the project show a viable approach to building scalable and easy-to-use software through the application of modern internet technologies and systematic development methodologies. The proposed solution is able to solve the problems related to the manual control of lifting operations by offering an efficient alternative that can boost productivity and improve communication with clients. In addition, there is potential for enhancing the system with additional functionality, including analysis and prediction capabilities.

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Dr. Mohamed Shakir Project Guide PSCS Presidency University	**Mr. Sreehari T M** Program Internship Coordinator PSCS Presidency University	**Dr. Md Ziaur Rahaman** School Internship Coordinator PSCS Presidency University

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ABBREVIATIONS

Abbreviation	Full Form
AMC	Annual Maintenance Contract
API	Application Programming Interface.
CSS	Cascading Style Sheets
EEG	Electroencephalogram
ERP	Enterprise Resource Planning
HTML	Hyper Text Markup Language
IDs	Identifiers
IoT	Internet of Things
IoTWF	Internet of Things World Forum
JS	Java Script
MUI	Material User Interface
PaaS	Platform as a Service
QA	Quality Assurance
QR	Quick Response (Code)
REST	REpresentational State Transfer.
SPA	Single page Applications
UI –	User Interface
UX	User Experience
AMC	Annual Maintenance Contract

Chapter 1

INTRODUCTION

Technological innovations have immensely changed how businesses function, handle information, and communicate with their customers. Currently, companies operating in fields like elevator servicing, telecommunication services, and management of infrastructure require an efficient platform that will aid in tracking assets, carrying out maintenance exercises, and ensuring proper communication with the customers. As such, the importance of an efficient system cannot be understated.

The purpose of this internship project was to develop a customer-based application within the PaaS environment. This project was undertaken at Axxonet Solutions and took six months (24 weeks). The platform used was run by Avia Enterprises, who offers efficient tools for handling databases, customers, and managing their operations as a whole. Utilizing modern web technologies, the platform allows for more efficient data processing and real-time interaction with customers.

Key Goal of Internship Program:
The key goal of the internship program was designing a Customer Application for the Lift Management System. The main target of such software includes being able to monitor and manage the assets owned by a customer, like lifts, including all necessary details as well as maintenance plans based on AMC. With the help of the Customer Application, users will be able to check upcoming services, change schedule of services, and review the service history of each lift. Moreover, customers will be able to generate service tokens via this application.

1.1 Background

The reliance on digital technologies has led organizations to adopt a more centralized approach in managing their operations, addressing customer queries, and processing requests for services. The conventional practices, which include manual recording and scheduling, have proven to be cumbersome, inefficient, and highly susceptible to human errors. Therefore, there is a need for an effective way to manage these tasks.

This is where the Platform as a Service (PaaS) framework has gained immense relevance. With PaaS, businesses can take advantage of an integrated platform without having to invest in their infrastructure, saving considerable time and costs.

Axxonet Solutions is known for providing digital solutions to help organizations streamline their asset management, customer relations, and service management processes. Using its proprietary platform called Avia Enterprises, it assists sectors like elevators management and telecommunications in optimizing their processes. Through the platform, companies can ensure that their assets and service processes remain organized within structured databases.

In this regard, the significance of designing an application that caters to the needs of its customers becomes quite apparent. Modern-day customers seek transparency, particularly regarding the assets they own. Making this information easily available to them becomes critical in enhancing their confidence and providing them a better experience.

The Lift Management Customer Application designed during this internship seeks to meet these requirements by making all the information related to the assets visible to the users. This includes asset information, ID numbers of the assets, and maintenance of assets as per the AMC agreement. Additionally, it facilitates monitoring of service timelines and initiation of service requests by the users whenever they desire.
In view of the foregoing scenario, the significance of modern-day applications in a PaaS framework can be realized very well.

1.2 Statistics

With the increased implementation of platforms in different sectors, there is a need for effective data management and customer-oriented applications. According to recent trends in the sector, above 80% of firms are opting for Platform-as-a-Service (PaaS) as they shift from using conventional software to cloud computing platforms. This enables firms to process huge amounts of data, automate important processes, and offer real-time services to their clients.

The sectors of elevators, telecommunications, and facility management are increasingly experiencing the need to undergo digitalization. This is primarily attributed to the complications associated with asset management and structured scheduling of services.

For sectors such as lifts and facilities management, firms have the task of supervising many assets and carrying out planned and timely maintenance of these assets. Studies have revealed that the implementation of scheduled and predictive maintenance programs could lower downtime by about 30 – 40%. Moreover, changing consumer expectations have seen more than 69% of consumers prefer using digital platforms to request services and track service progress.

1.3 Prior Existing Technologies

Prior to the introduction of PaaS, many companies utilized legacy approaches involving manual recording, spreadsheets, and stand-alone computers to store asset and service data. Besides being cumbersome and labor-intensive, these techniques did not facilitate any mechanism for managing updates, monitoring maintenance, or responding to customer requests in an efficient manner. Thus, companies, including those operating within the elevator service and telecom sectors, had to experience delayed service delivery and inefficient asset management due to the lack of a proper system. In addition, there were high chances of error.

With the advent of new technologies, the outdated approach was increasingly substituted by web applications and cloud solutions. Platforms like Avia Enterprises provided companies with a unified solution where data could be accessed, stored, and managed at all times. The benefits associated with such platforms included enhanced efficiency through automation and management tools such as customer management systems and scheduling. In addition, frontend

frameworks and API-driven systems were adopted in application development.

Apart from offering solutions to business organizations, Axxonet Solutions has also innovated in certain areas such as the technology used in medicine. For instance, one such innovation includes developing high-tech platforms such as the SuperEEG system, which makes use of EEG technology. This demonstrates the ability of the organization to develop intricate applications and conduct technological research.

This kind of knowledge can be utilized in other sectors such as asset and service management to create systems that not only are efficient but reliable as well. The move from the old technological ways of doing things to using digitized systems showcases how innovation plays a key role in enhancing performance.

1.4 Proposed Approach

The main objective of this internship includes development of customer-oriented web application for lift management in PaaS model offered by Avia Enterprises. The application is intended to provide customers access to all necessary information regarding lifts and their respective maintenance plans.

This platform provides all details about asset ownership and management to the customers. These details include unique ID number and specifications about each lift in possession of the customer. The customer can use this web application to keep a track on schedule of AMC plan along with future service plans and overall timeline of services provided to the lift.

Technically, this application is developed using technologies such as React for frontend development in addition to HTML, CSS, and JavaScript. The concept of components along with state management and APIs play a significant role in developing this web application. Frontend is developed to communicate with the backend through API calls to fetch real-time data stored in the database through an interactive interface.

Moreover, apart from providing information on the asset itself, this system can also help to increase interaction with the client through certain features, including re-scheduling of service appointments as well as requesting service via tokens.

The overall system is consistent with the goals of Axxonet Solutions as it provides an innovative solution to asset management problems.

1.5 Objectives

In addition, the main purpose of the internship is to create a web-based application for lift management that would be beneficial for customers, being centered on them. In particular, such an application should be developed to achieve higher transparency, efficiency, and customer satisfaction through access to the information about assets and services. The following goals should be met:

1. Creating an Effective and User-Friendly Customer Interface

The first major goal to meet in this internship is to create a convenient and easy-to-use interface for customers. It is supposed that with such an interface, customers can easily use the features provided by the application and navigate through them. In particular, customers will be able to easily view asset information, look at maintenance schedules and make service requests. The interface will allow using various design solutions for the layout that will promote easier usage of the application.

Modern web development technologies (React, for instance) allow creating reusable and highly efficient interfaces, which can easily adapt to increasing volumes of data.

2. Simplifying Asset Management for Users

The other main aim of the app is that of ensuring that customers are able to gain insight into their

lift assets. In the application, users can find all relevant information regarding their assets such as asset IDs, locations, and even service records. By having all such information stored at one place in the system, the hassle of keeping track manually and reliance on the service provider for basic information has been completely eliminated.

By using the APIs of the backend system and thus enabling real-time retrieval of data, it is ensured that the customer is always updated with the latest information pertaining to his asset.

3. AMC-Based Maintenance Monitoring

Lift maintenance is an essential factor in lift systems, and through this project, we seek to make it easy by implementing maintenance tracking using AMC plans. This will allow our clients to check their service schedule, the next date of service, and previous service records systematically.

Through this goal, we guarantee that our clients will be up-to-date on their service schedule and can plan effectively for their operations. With automation, we minimize the risk of missing any scheduled services, improving the performance and efficiency of the system.

4. Manage Service Scheduling and Rescheduling Efficiently

This solution will facilitate the ability for customers to be able to manage their appointments successfully. The application users can view their future schedules and reschedule them according to their convenience.

Implementing scheduling functions will entail working with date and time information and validating data input in addition to integrating the application via an API call. The goal of this function will be to minimize delays and maximize efficiency in meeting customer needs.

5. Service Token Generation and Management System

A significant goal of the project is to come up with an efficient system that enables customers to

generate service request tokens or issues related to their lifts. This mechanism helps ensure that there is an orderly process of reporting any problems, and users can be able to monitor progress and receive feedback from the service provider.

The introduction of this aspect makes it easy to manage customer requests since it helps create accountability on the part of the service provider. Therefore, there will be fewer complaints from customers due to non-resolution of issues.

6. Build Real-World Skills in Full-Stack Development

Apart from being able to implement various aspects of this project, another important goal of this internship is getting practical training in full stack development in a professional setting at Axxonet Solutions. It involves learning to work with frontend tools, integrating APIs, managing application states, and working in a team of developers.

During the process of learning, skills related to software development life cycle, debugging, and other problem-solving methods are learned. Learning about industry standards and practices helps in laying a foundation for future career opportunities.

1.6 Sustainable Development Goals (SDGs)

The customer application designed for lift management systems correlates well with numerous United Nations Sustainable Development Goals (SDGs). The primary reason for this correlation is the fact that the software emphasizes the efficient use of resources, sustainable infrastructure development, and improved service delivery by means of digital innovation. Through the use of advanced technology, Axxonet Solutions, via its Avia Enterprises platform, provides enhanced service delivery solutions.

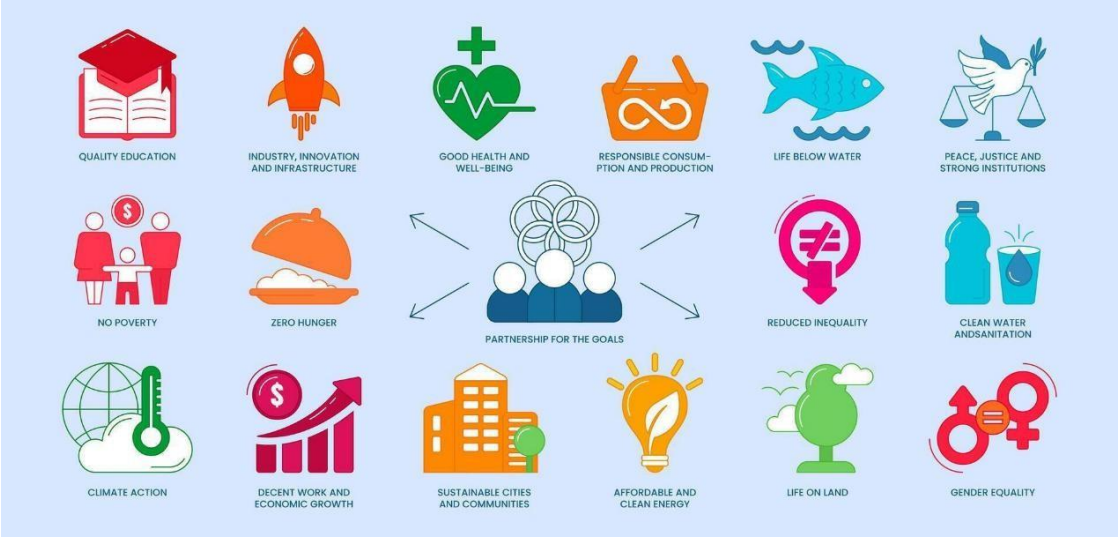

Fig 1.1 Sustainable development goals

SDG 9: Industry, Innovation, and Infrastructure

In summary, the current research project is in line with the goals of the SDG 9 because it uses novel technology to optimize industrial practices and manage infrastructures effectively. For example, developing the user interface will allow the lift management system to be managed digitally, thereby eliminating the need for physical records.

Additionally, using modern web technologies and APIs for service management will enhance service delivery. Finally, adopting the PaaS model will ensure that organizations can develop and implement novel solutions without relying on significant physical infrastructure investments.

SDG 11: Sustainable Cities and Communities

Lifts are an integral part of urban infrastructure. They are necessary for ensuring the smooth running of high-rise buildings, smart cities, and other facilities. To ensure proper functioning of this equipment, it needs to be maintained properly and serviced periodically. Customer app development can relate to SDG 11 as it can assist in better management of the lift system.

Service planning, monitoring of maintenance work, and obtaining information about the state of

the lift system can play a significant role in preventing equipment malfunctions.

SDG 12: Sustainable Consumption and Production

Through such software, responsible use of resources is facilitated through good maintenance planning and prevention of any service that is not required. Through planning for services according to the AMC schedule, organizations will be able to make the best out of their processes while at the same time eliminating wastage of valuable resources.

Another benefit of this technology includes reduced reliance on paper documentation since all documentation is done using the computerized system. This ensures good quality of service delivery without compromising on responsible consumption and production.

SDG 8: Decent Work and Economic Growth

The implementation of such technological advances like the customer app will lead to increased efficiency, which is also an opportunity for the creation of skilled jobs in the technology industry. Increased efficiency is essential because it ensures efficient service delivery and overall business productivity.

Furthermore, doing an internship at Axxonet Solutions can be useful because one gains knowledge about full-stack development, which prepares one for future employment in the IT industry.

1.7 Project Report Outline

The structure of the project report includes several chapters, which concentrate on particular issues of the application development.

The project report starts with the Introduction chapter, which includes information about the

general characteristics of the project. It describes its background, goals, and the problem the project intends to solve by providing a necessary technological solution for managing the lift services.

The second chapter is called Literature review and represents a comprehensive study of modern technologies, frameworks, and literature on the topic. The chapter discusses various methods of web applications development such as REST or GraphQL APIs, and frontend or system design options.

Methodology describes in detail the process of working on the project and covers all stages of the development from requirement analysis and system design up to its implementation and testing.

Project Management

This chapter provides information on planning and executing the project. The chapter is based on the internship schedule, task distribution, and risk evaluation. It provides insights into the management of the project within the six months and measures taken to overcome any difficulties that were encountered during the project.

Analysis and Design

In this chapter, the requirements of the system and its architectural design are explained. This includes block diagrams, flowcharts, and other models, such as the domain model, communication model, and functional view of the system.

Hardware, Software, and Simulation

In this chapter, the hardware and software environment used in the project are outlined. It covers the hardware and software requirements and the simulations performed during the project.

The chapter on Testing and Validation highlights the methods of testing the application system. Issues regarding test case analysis, test plan and test results are highlighted in this chapter as a means of testing performance of the system.

The social, ethical, legal, sustainability and safety aspect chapter looks at the impact of the whole application development process.

Finally, the chapter on conclusion is included as a means of summarizing the findings from the whole project. Possible improvements for future projects are also discussed in this chapter. Additionally, references and base papers are included as part of the report for all the sources used.

Chapter 2

LITERATURE REVIEW

This segment discusses relevant literature, discoveries, and methods to lay the groundwork for the subject matter, especially on the issues of asset and service management. In light of the fast-evolving technologies, companies are resorting to cloud technology and web application development in order to maximize efficiency and provide better service to their customers. Several techniques, tools, and technologies have been investigated by scholars in this field.

As far as this internship with Axxonet Solutions is concerned, the development of a customer application for lift management would require the use of several technological fields such as front-end frameworks, back-end architecture, cloud computing, and user interface/user experience design. The Avia Enterprises system illustrates that centralization systems can be used to manage company data and customer interactions effectively.

In addition to that, the need for real-time access and transparency within the service industry has contributed to the increased use of APIs and dynamic UIs. The use of development methodologies like component-based design and state management has enhanced the efficiency of applications. All things considered, this paper offers a good basis of knowledge about current research and technologies, helping gain an insight into the methods employed in the proposed system.

2.1 Introduction to Web Application Development

The development of web applications has seen a lot of growth in the last ten years because of the rising need for scalable and interactive web applications. Today's web applications have transformed from simple websites into more complex applications that incorporate both frontend and backend components, along with databases to provide real-time functionality.

Definition of Web Application Architecture is the architectural design of a web application, which defines the way various components like the user interface, APIs, and database communicate among themselves. It can be considered a design pattern that directs the development of effective applications. As per recent research findings, an optimal architecture makes application more efficient, increases user satisfaction, and ensures the future maintainability of applications by defining the interaction between various parts of the application.

When considering the internship project, it is clear that the development of the customer application for lift management utilizes concepts from modern web applications. Use of front-end framework, API, and cloud platform are some of the key characteristics of today's web applications.

2.2 Web Application Architecture Evolution

Web application architectures have evolved from basic client-server approaches to more sophisticated structures like multi-tier and service-oriented architectures. The client-server approach was employed in early applications, which involved user interface operations on the client side and data processing on the server side. Nonetheless, due to increased complexity, this approach was inadequate in meeting the demands of scalability and performance.

Newer architecture styles like multilayered architecture and service-oriented architecture (SOA) break up an application into several layers like presentation layer, business logic layer, and data layer. The benefit of this kind of architecture is higher maintenance and the ability to develop each layer independently. Studies indicate that such architectures can easily cope with evolving business needs.

Moreover, the changeover from monolithic architecture to modular architecture has had a great effect on the domain of web development. Monolithic applications used earlier were highly coupled and difficult to maintain and scale up, whereas modern modular architecture enables us to create our applications through independent modules.

Table 2.1 Comparison Table

Criteria	Traditional Systems	Modern PaaS-Based Systems (Avia Enterprises)
Data Management	Data is managed manually or kept in separate systems	Data is managed in one centralized cloud system
Accessibility	Not widely available and only access in specific locations	is available at any time and place through the internet
Scalability	Troublesome when handling more data	Easy to scale because of cloud architecture
Maintenance Tracking	Manual maintenance tracking via paper documentation and spreadsheets	Automatic maintenance tracking with real-time information synchronization
User Interaction	Low interaction with the user or zero interaction	High interaction and usability
Real-Time Updates	Update data is not easy and often delayed	Real-time update information and synchronization
System Integration	Low or no integration with other systems	Highly integrated with other software by application programming interfaces
Performance	Lower speed due to the manual process	Higher speed because of optimization
Reliability	Error-prone and lacks backup and recovery mechanisms	Highly reliable with backup and recovery strategies
Cost Efficiency	Higher costs in the long term	Lower costs as a result of cloud resources
Security	Below standard security or	High security due to

	lack thereof	advanced techniques such as encryption
Collaboration	Challenging to collaborate between several users	Real-time collaboration between several users
Flexibility	Rigid and not easy to change	Flexible and easy to upgrade using advanced technologies such as React
User Experience	Poor user experience or no usability experience	Improved user experience and user interface/user experience design

2.3 Platform as a Service (PaaS)

With the rise in development needs, there has come an evolution of Platform as a Service (PaaS), which is considered a major component of modern-day web application development. With PaaS, developers have access to a cloud-based platform that supports software application development, testing, deployment, and management without having to be concerned about the backend server and storage.

This is because of the presence of tools for developing, collaborating, and managing databases all within the PaaS platform, enabling several developers to simultaneously collaborate on the same software product development process.

However, there are also difficulties associated with developing software applications for PaaS. Cloud environment variation and many different services make the architecture decision difficult. According to some studies, assessing the application prototype in cloud environments contributes to the improvement of the system.

The adoption of the PaaS solution offered by Avia Enterprises fits these ideas, contributing to the effective development and deployment of the customer application.

2.4 Importance of APIs in Web Application

An API is an important component in web applications where it facilitates communication among various software applications. An API allows the frontend applications to communicate with other services from backend and external sources, which makes the API important for dynamic web application that uses databases to deliver content.

Modern research suggests that the importance of APIs lies in their capacity to facilitate interactions between different software services and thus improve modularity and scalability.

Moreover, the role of APIs becomes indispensable for microservice architecture design where applications are segmented into small units called services. The services then interact using APIs to develop and implement the components independently. It results in improved agility and faster development cycles.

On the other hand, integrating APIs faces several challenges like security issues, versioning of APIs, and handling errors. Researchers have found that mishandling of API keys and ignorance on the part of developers result in problems related to API security.

In the developed application, APIs were implemented to obtain details regarding assets, maintenance schedules, and services.

2.5 Frontend Technologies and User Interface Design

The role of frontend development is vital in ensuring that web applications are effective in use. The introduction of frontend technologies like React has brought about a major change in the design of user interfaces through component-based designs and state management techniques.

Single page Applications (SPA) is increasingly popular because of its capability to update the user interface dynamically without refreshing the whole page. This makes the application more efficient, responsive, and fast. Research shows that SPAs improve efficiency and lighten the workload of servers.

The UI and UX designs are also very important for frontend development. A good user interface increases accessibility and minimizes mistakes, leading to greater satisfaction on behalf of the

users. When dealing with customer applications, ease of use and navigability are vital for ensuring successful communication.

As regards my internship, utilizing React and proper UI principles will help guarantee that the customer application will be both responsive and scalable.

2.6 Microservices and Contemporary Application Design

The microservices approach is a major leap forward in terms of application design, as it involves splitting applications into microservices. Every microservice is responsible for a certain activity and exchanges data with other microservices via their APIs.

Some of the benefits of using the microservices architecture include scalability, flexibility, and isolation of faults. In addition, it allows organizations to independently update microservices.

Unlike monolithic systems, where components have more interdependencies and need to coordinate their work more, microservices minimize dependencies and enable independent work by different parties. At the same time, microservices bring problems such as greater complexity in terms of managing APIs, service communication, and monitoring.

With microservices architecture, apps like the lift management customer app will ensure that separate functionalities can evolve independently from one another.

2.7 Applications with Customer Orientation

Today's applications are customer-oriented in nature and provide customers with real-time data and services. These applications help in achieving greater transparency and quick response.

In service-based sectors like lift management, the customer is required to have access to his/her asset information and service details. With digital apps, the customer can track his/her asset and engage with the service provider.

Research shows that customer-oriented systems increase efficiency and improve the customer experience since it eliminates manual efforts and offers real-time interaction. Customer-oriented systems also allow building trust and strengthening customer relationships.

The customer application, which was designed during the internship, adheres to these concepts as it has several features, including asset tracking, AMC monitoring, rebooking of services, and token generation.

2.8 Summary of Literature Review

The significance of modern web technologies in making efficient and performing applications has been identified in the literature review. Modern architectures, the use of PaaS services, and APIs have brought a huge revolution in the process of application development.

React on the front end side, along with the microservice architecture of the applications, has helped in making applications more flexible and efficient. The focus on customer-centered design has ensured the provision of the right solution according to the customer requirements.

In totality, the ideas presented above in the literature review have provided a good theoretical base for the application development undertaken at Axxonet Solutions during my internship period.

Chapter 3

METHODOLOGY

For the current project, the chosen methodology is a structured and systematic method of developing a customer-oriented application for managing lifts within the PaaS architecture offered by Avia Enterprises. It entails creating the front-end and back-end aspects of the application and combining the two into one efficient system. An iterative development technique will be used whereby each aspect of the application is designed, implemented, and tested individually.

Front-end design is achieved through React, a very popular JavaScript library used for developing user interface (UI) components. Furthermore, MUI (Material UI), which comprises components designed using the principles of material design, will also be used. Each feature, such as asset management, maintenance management, and request management, will be implemented separately using individual components.

The frontend and backend are connected via APIs, facilitating data transfer at real time. State management and data processing are performed effectively via React hooks including useState and useEffect. The users will be able to see their latest updates concerning lift asset, maintenance and service booking without any manual refreshing of the website. In addition, APIs enable the system to effectively manage big amounts of data.

Testing plays a vital role throughout the development process to improve the reliability and performance of the app. The functional testing will be conducted to test each module for usability and responsiveness and UI/UX testing for better user experience.

3.1 Requirement Analysis

Requirement analysis is one of the most vital stages of creating an application as it aims to

establish the requirements of both the organization as well as its end users. In this phase, extensive discussion and observation sessions were carried out regarding the methods adopted by various industries, especially those offering lift management services, to manage their assets, schedule maintenance operations, and interact with their customers. Some of the shortcomings observed during this stage include the lack of transparency, inability to access data instantly, and ineffective communication between customers and the organization.

One of the key requirements during the requirement analysis was the need to understand the importance of the PaaS Platform provided by Avia Enterprises in order to understand the mechanism through which organizations use their platform to store all the relevant details pertaining to assets and services. All the details about assets, customers, and services were stored within the platform, and hence the software had to be developed in such a manner that it would integrate itself seamlessly into the system. The APIs and data structures for pulling out details like ID, AMC schedule, and timelines for the services were needed.

From a user standpoint, the requirements were geared towards ensuring that the user gets a very simple and user-friendly interface which will enable them to access their data without much hustle. Some of the functional requirements of the application include the ability for the users to view their assets, check their maintenance schedules, check their future services, reschedule service meetings, and initiate service requests (tokens). Non-functional requirements of the project include security, performance, and usability. The frontend used React and MUI frameworks.

In addition, the requirements analysis stage entailed looking at the probable problems and restrictions, such as managing huge amounts of data, providing real-time information, and guaranteeing data accuracy. The issues of error handling, API failure, and user input validation were taken into account so as to make sure the app was reliable. The requirements analysis stage provided a great foundation for the system design stage through proper identification and analysis of both the functional and non-functional requirements.

3.2 System Design and Architecture

The design and architecture of the customer application have been designed to ensure scalability,

modularity, and smooth handling of data in the PaaS environment of Avia Enterprises. The design of the application utilizes a layered architecture comprising three main layers including the frontend, API communication layer, and backend data services layer. These three layers operate independently but work seamlessly to deliver all functionalities expected of the application. The design ensures that the application can handle dynamically changing data, multiple users, and other tasks associated with the management of assets and services.

The frontend layer of the application uses the React framework together with the Material User Interface (MUI). The application uses a component-oriented architecture whereby the various parts of the application, including asset details, maintenance, service, and token generation, are modeled as reusable components. This helps in ensuring maintainability of the application and ease of modifications or upgrades when necessary. MUI has been used in this regard because it offers pre-built UI controls hence fast development and user-friendly interfaces.

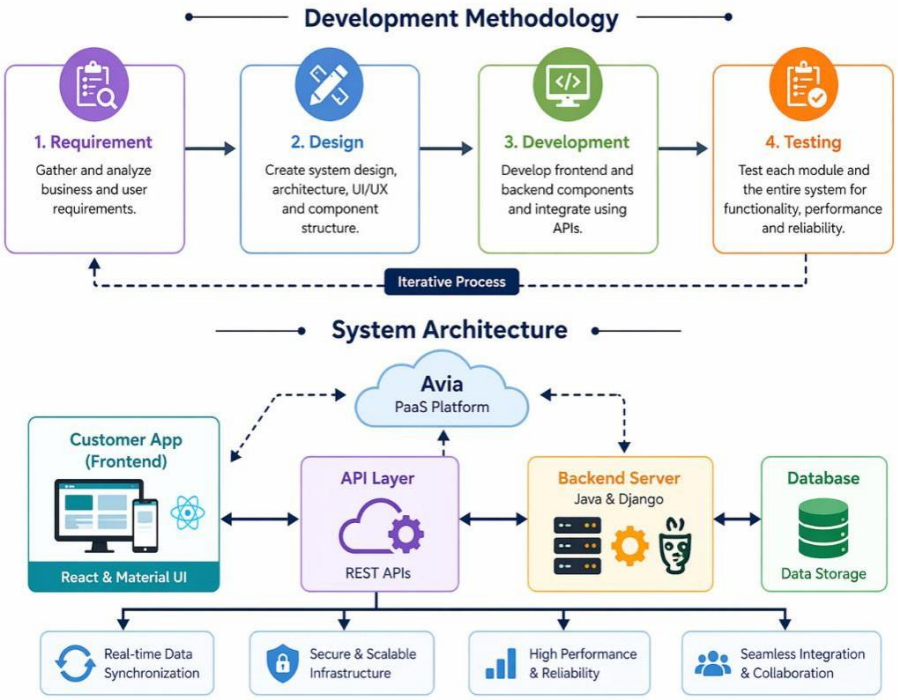

Fig 3.1 System Architecture

Data exchange between the frontend and the backend occurs through API integration. The RESTful APIs will enable the retrieval, updating, and management of data associated with customer assets, AMC plans, scheduling of services, and service requests. React hooks like

useState and useEffect are implemented for state management and processing asynchronous data in order to ensure that changes happen in real time. In turn, this ensures that the application stays dynamic, as users can see changes as soon as they are made in real time.

From the backend perspective, the application utilizes the centralized nature of data management in the platform in order to ensure the storage and organization of all necessary information including the asset records, customer information, and services. The architecture is designed with data consistency, security, and information retrieval in mind. Furthermore, the implementation of this architecture includes various methods of error handling, validation, and performance optimization. With the use of the above-mentioned architecture, the application helps to achieve the goals of Axxonet Solutions.

3.3 Frontend Development using React and MUI

In order to develop the frontend of an application, the modern technological stack includes React and Material UI (MUI) framework. The implementation allows for creating dynamic and responsive interfaces that are user-friendly and easy to navigate. React framework helps in splitting the interface into several independent components that can be used separately. In particular, such components include dashboards, assets' detailed information, schedule of services, and service request tokens among others. Thanks to modularity, each component can be easily updated and improved without interfering with the overall system.

Moreover, using React allows for developing the app that is easier to maintain since each of the components is responsible for a certain task or action. The Material UI (MUI) framework helps in enhancing the interface by incorporating a number of components that may be useful in the process of app building. They include buttons, tables, forms, dialogs, and navigation among others. Thus, thanks to pre-built components, the development process becomes easier and faster. Additionally, thanks to the framework, the application becomes fully responsive, and its interface can be consistently styled.

In addition to UI design, hooks such as useState and useEffect in React are essential for controlling the functionality of applications. While useState hook is required for managing the state in components, the useEffect hook is necessary for dealing with side effects such as API

calls. They allow dynamic content rendering so that users always get updates on information about their lift assets, maintenance scheduling, and services. With the use of data from APIs, the application makes the communication process convenient and effective for users.

3.4 State Management and React Hooks

State management is essential in building the customer application, as it helps in ensuring efficient data handling and synchronization between the UI and information underneath. In this project, the concept of state management is utilized through the React built-in hooks feature without the need for additional library implementations. The app will manage several types of data from different sources, such as lift asset details, maintenance, servicing schedule, as well as service requests generated by the end-users. Through effective state management, all data flow processes in different parts of the UI can be done effectively.

One of the hooks in the React framework is the useState. This feature is mainly used to manage the dynamic state within the components. In this case, the state variables help in storing data related to user input information, API responses, and other UI-related variables, including loading and error status. As an illustration, in cases where users view their lift asset or send service requests, the data collected from such actions is saved in state variables, which get updated in real time and displayed on the screen.

Effect hook is utilized to perform side effect operations like fetching data from APIs, rendering UI based on change, or making other operations upon mounting or re-rendering of components. It becomes extremely crucial in order to fetch data in real time with respect to asset data, AMC scheduling of maintenance tasks, and service requests. With the help of useEffect, data will get loaded automatically, and any updates in the data will be reflected without fail. The two hooks make an effective combination that would facilitate efficient management of states and side effects for building a responsive and interactive application as desired by Axxonet Solutions.

Chapter 4

PROJECT MANAGEMENT

Project management was one of the vital aspects in facilitating the successful implementation of this internship project. Project management was aimed at organizing, planning, and overseeing all activities related to creating a customer application intended for lift management. This internship lasted six months, which can be counted as 24 weeks, and within this period of time, all activities were carried out in accordance with a structured plan. This means that all activities in developing the customer application took place through several stages such as requirement analysis, design, development, testing, and improvement of the application.

Another important aspect of this project is that its implementation was based on effective time management. All activities were prioritized depending on their importance; weekly goals were set to track the progress during the entire development process. Besides, close communication with the mentors was an important aspect to facilitate the progress.

What is more, it is necessary to mention that this internship project was implemented with high flexibility. With the progress of development, feedback was regularly received and taken into account for improving the customer application. Overall, this internship project was successfully implemented according to a structured plan with effective time management and regular feedback from mentors.

4.1 Internship Timeline

Axxonet Solutions' internship program was conducted throughout a span of six months, starting from January 5th to July 4th and was organized in a manner that allowed progressive learning through step-by-step process of learning and implementing full stack developer skills. A timeframe of one month was allotted to different learning areas as well as completing milestone

projects in order to achieve a seamless progression from basic concepts to applying what is learned into the process of actual development work.

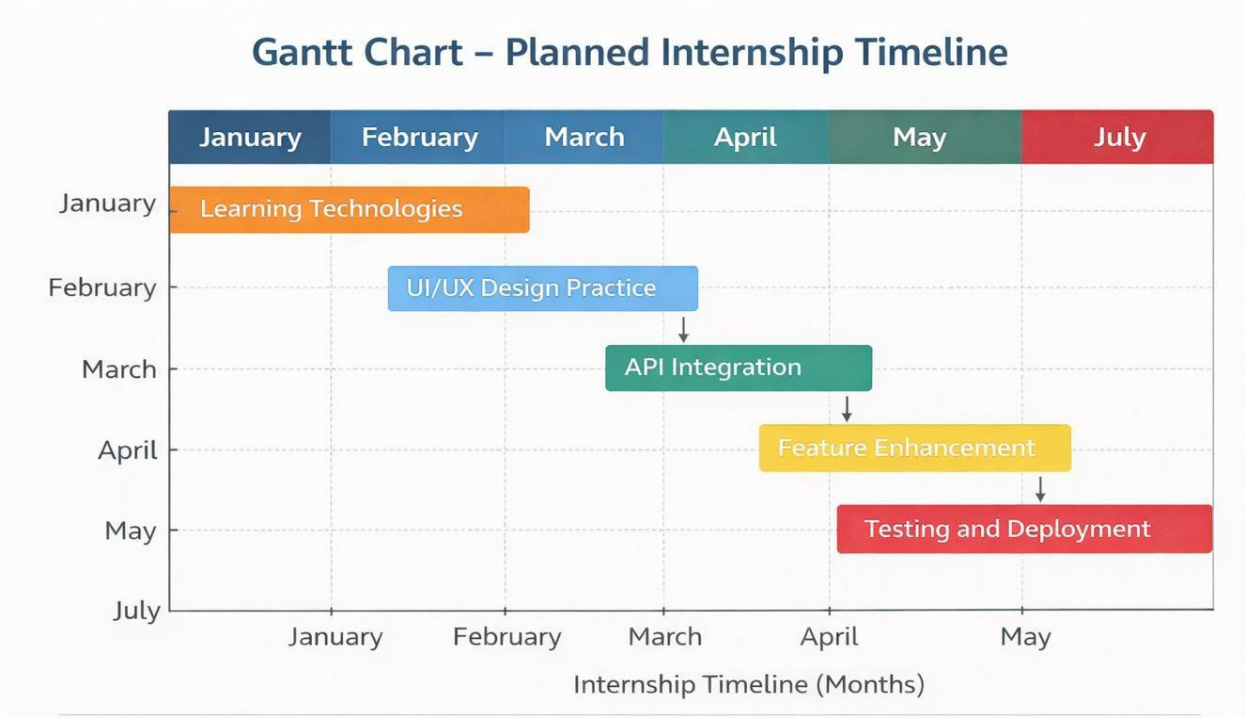

Fig 4.1 Internship Timeline- Gantt Chart

In the **first month** of the internship program, emphasis was laid on the acquisition of fundamental skills by acquiring skills related to various programming and web development languages such as HTML, CSS, Javascript, React, Java, Python, and Django. This was important in laying a solid foundation of knowledge in terms of both frontend and backend as well as gaining an introduction to the technologies used in developing a complete web stack.

In the **second month**, the concentration shifted to utilizing the acquired knowledge through the creation of personal websites. In this regard, the process involved the utilization of the gained frontend skills in implementing layouts, designs, and interactions among others. It offered opportunities for enhancing problem-solving skills as well as developing confidence in developing web pages. The stage marked the transition from theory to practice.

In the **third month**, the start of the major project began, and there was a need to initiate work on the client application for managing lifts in the Avia Enterprises setting. It was necessary to

concentrate on comprehending the needs of the project as well as creating the user interface for the site. The stage involved initiating the designs of various pages that will form the site, especially in relation to assets, services, and maintenance.

In the **fourth month**, development made progress with the introduction of navigation and API integration. The app was becoming more functional with the connections being made among various pages as data became available from the backend systems. This stage proved to be critical in learning about real-time data manipulation, state management, and backend and frontend interactions. Testing and design were also done throughout the process to improve on the user experience with regard to user interface (UI).

The next milestone during the internship will be the **fifth month** during which the development process of the customer app will involve more improvements. The main objective of the month is the improvement of the functionalities of the app. Various improvements will include the ability to manage services such as service token management, as well as more improved timelines for the various services. The main emphasis will also be placed on API integration and the handling of edge cases, especially in data manipulation. Further enhancements will be made on the front-end using the React framework and Material UI to ensure consistency and good functionality.

The **sixth month** will primarily focus on finalizing the application and preparing it for deployment within the Avia Enterprises environment. This phase will include comprehensive testing, performance optimization, and validation of all implemented features to ensure they meet the required standards. Documentation of the project, including technical details and user guidelines, will also be completed during this stage. Additionally, feedback from mentors and team members at Axxonet Solutions will be incorporated to make final improvements. The internship will conclude with the successful completion of the customer application, demonstrating the practical application of full-stack development skills and the ability to deliver a real-world, user-focused solution.

4.2 Risk Analysis

Risk analysis plays a crucial role in the process of project management since it enables the identification of any possible obstacles in order to avoid problems during the implementation and

launch of the application. In the process of creating the customer application for lift management within my internship at Axxonet Solutions, there were several risks that were analyzed regarding the project's implementation. They include performance risk, data risk, API risk, and UX risk.

One of the key issues with the project involves API integration and data consistency. Given that the system makes use of real-time data obtained from the backend services in the Avia Enterprises, any problems in terms of response from the APIs, delay, or any inaccurate data could lead to problems in terms of performance of the application. This could be due to latency in networks, downtime on the servers, or wrong data formats. However, there are some measures put in place to avoid any problems with the application in case there is an issue with the API.

Other possible problems include performance issues arising out of frontend management of states in the application. Since there are many components in the application such as assets, scheduling, and service requests, poor management of the states may lead to performance issues and slower rendering of information on the frontend. Implementation of efficient tools such as React along with the use of effective hooks such as useState and useEffect will be helpful in mitigating these potential problems. However, if the hooks are used improperly, it may lead to unnecessary rendering and ineffective memory utilization.

In addition to this, it is also important to consider risks related to the users and operations associated with the application. This includes issues related to the ease of use of the application, wrong input from the user, and unfamiliarity with the interface. If the application is not easy to use, there is a possibility that it will not be used much and the users may feel dissatisfied with the product. Other risks relate to security issues that arise due to unauthorized access to information stored on the computer. This can affect the reliability of the application.

4.3 Internship Budget

In the case of the internship conducted at Axxonet Solutions, there were no direct monetary spendings, since the majority of the costs that might be incurred through the development process were already covered by the organization. As the project entailed the creation of software in a PaaS structure (such as that of Avia Enterprises), very little investment in any hardware was needed. Most of the platforms and tools necessary for the work process were open-source or free

of cost and were already available in-house.

The main elements of the internship budget are represented by the indirect costs of system requirements, internet connectivity, and educational resources. A computer or a laptop with proper configuration would be needed to operate development frameworks and tools like React.js and local servers. At the same time, an internet connection would be required to use the cloud, APIs, and documentation. Educational resources could also be used during the course of the development process, yet most of them were already free.

One more important feature that has to be considered is time and efforts, which are extremely valuable for any internship. For instance, it took me 6 months (24 weeks) to study new technologies, implement some additional features, debug the code, and finally refine the software. Although there were no substantial expenditures, it was very important to dedicate my time and efforts to the internship.

Finally, it should be noted that this internship budget is minimal since the organization possesses necessary resources. Besides, all software used during this internship is based on open-source technologies. As a result, the main thing for interns is to gain knowledge, get practical experience, and complete the project successfully.

Chapter 5

Analysis and Design - Introduction

The analysis and design stage is one of the crucial stages of a development process, as it provides the basis for creating a good architectural structure of a software product. In other words, this stage is focused on analyzing system requirements and creating an efficient architecture based on those requirements. During my internship at Axxonet Solutions, the analysis and design stage was particularly important, as far as it helped shape the development process of the customer application.

The analysis stage is aimed at analyzing system requirements, both functional and non-functional, as well as at analyzing data flow and core modules of the application. For example, it is required to understand which modules will allow us to track assets, plan maintenance tasks, and manage service requests. By analyzing all of these aspects, we will be able to develop the software without errors and integrate it into the PaaS offered by Avia Enterprises.

The design stage will rely on the findings of this analysis by converting the gathered requirements to a system design. The system design will involve definition of system architecture, definition of modules, creation of diagrams and data structure designs among others. In addition, a modular strategy will be employed for the system whereby the system will be segmented into modules like the front end, API, back end and database. In doing so, the system will become more scalable and manageable while at the same time making the use of modern technologies to ensure an intuitive UI.

In essence, the analysis and design process guarantees an organized, focused and efficient development cycle.

5.1 Requirements

The requirements for the application to be used by the customer are determined according to the needs of both the organization as well as the end-users, so that the system meets the objectives of lifting management successfully. In the internship with Axxonet Solutions, during the phase of requirement analysis, the interaction between customers and the service providers was taken into consideration. The application is developed to facilitate customers to access information about their assets, monitor maintenance actions as well as communicate service requirements to the provider using an integrated system along with Avia Enterprises.

The requirements are stated in terms of functionality, which includes the features that should be incorporated in the application. It includes features like displaying customer asset details, which may include the details regarding lifts as well as the lift ID. In addition, the details of AMC (annual maintenance contract) of the customer as well as maintenance schedules according to the contracts must be provided. Service timelines will be displayed that will include service activity performed in the past as well as future ones. Customers can request service tokens as well as reschedule future service activities.

Non-functional requirements are concentrated on such issues as system quality and performance. The software should be available from various devices, from desktop computers to mobile devices. Another important aspect to consider is system performance when there are many users and large amounts of data involved into processes. Customer data should be protected, which requires appropriate authentication and data protection. At the same time, usability and reliability of the product are necessary for convenient interaction with the system.

In addition to functional and non-functional requirements, certain integration requirements can also be mentioned. The product requires integration capabilities in terms of exchanging data through APIs with backend services. These include access to data concerning assets and their maintenance, as well as the requests related to services and support of customers. It helps to create an effective connection and provide timely and up-to-date information regarding

customers' demands

Overall, defining the list of requirements allows creating a clear framework for further development of the system.

5.2 Block Diagram

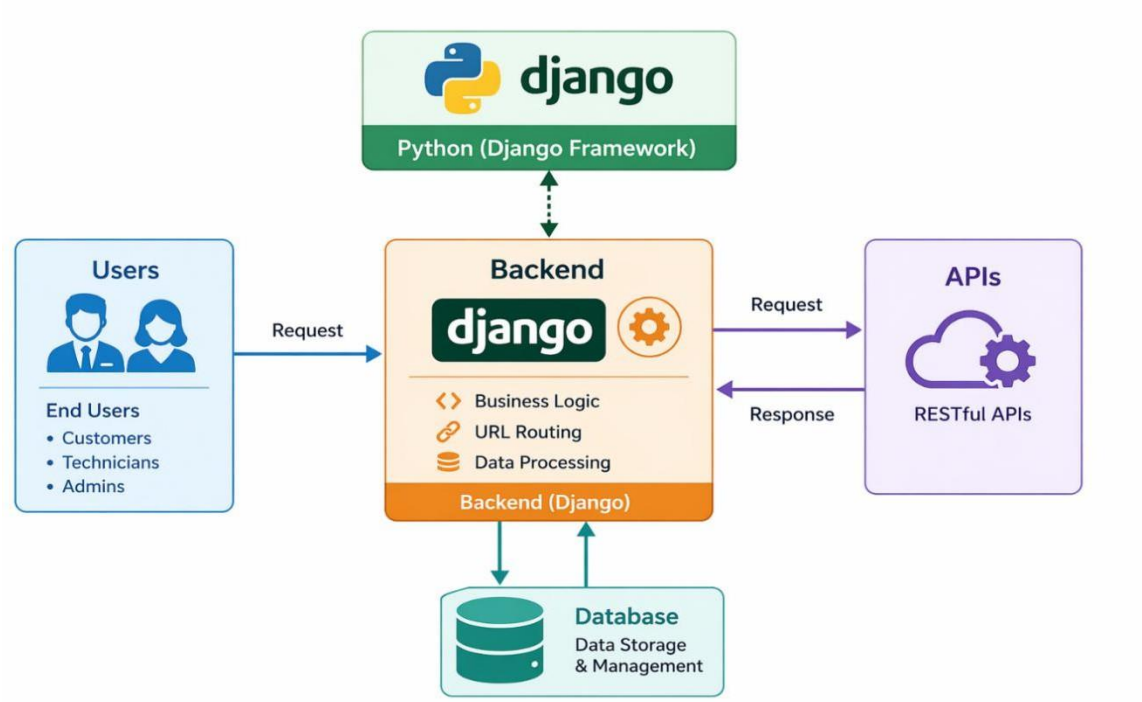

Figure 5.1: Customer Application Architecture

Figure 5.1 shows the customer application architecture that was developed based on Python and Django. It shows the architecture of the application and describes the way various parts of the application interact with each other when responding to the user requests in order to return the information required by the user.

The Users block corresponds to the end users of the application, which mainly represent the customers of the system that use the software in order to view the list of their assets, check the maintenance schedule and make service requests. Users perform actions like logging in into the system, requesting asset information and making service requests. In general, the Users block

corresponds to the starting point of the system and all the activity starts there.

The Backend (Django) block is the backbone of the system. This part contains the whole logic of the system. Using Python and Django, it performs all the activities related to the processing of the requests initiated by the user. This includes such processes as validation and storage of service requests, data authentication and accessing the database.

API Layer (handled internally by Django) acts as a layer between the backend and any outside or inside services. It handles the user requests which are passed into the system, processes them according to the rules in the backend code and generates responses accordingly. APIs make sure that data handling and transfer is efficient, streamlined and secure, making sure that real-time updates happen in the application.

Database block is responsible for the system that stores all the data related to the application. This can be customer data, data regarding lifts assets, AMC, service timelines, service requests etc. This interaction happens between the backend and database, allowing for storing/retrieving data as needed by the application.

In essence, what the above block diagram shows is how there is a clear way in which the data flows from the user to backend code in Django, is stored in the database and then is returned back to the user in response.

5.3 System Flow Chart

System flow charts depict the entire process flow of the application from start to end as depicted by Figure 5.2. In this system flow chart, it becomes evident how user input is processed by the system right from login to completion of service request. The flow chart is important since it provides the necessary systematic flow of operations to ensure efficiency in data processing and communications within the application.

The first step in this process is user login where the user gets into the application successfully and the system retrieves personal information from the backend of the application through API. In this process, the data of the customer is accessed including lift assets and maintenance schedule of the lifts. This process is represented clearly in Figure 5.2 as follows:

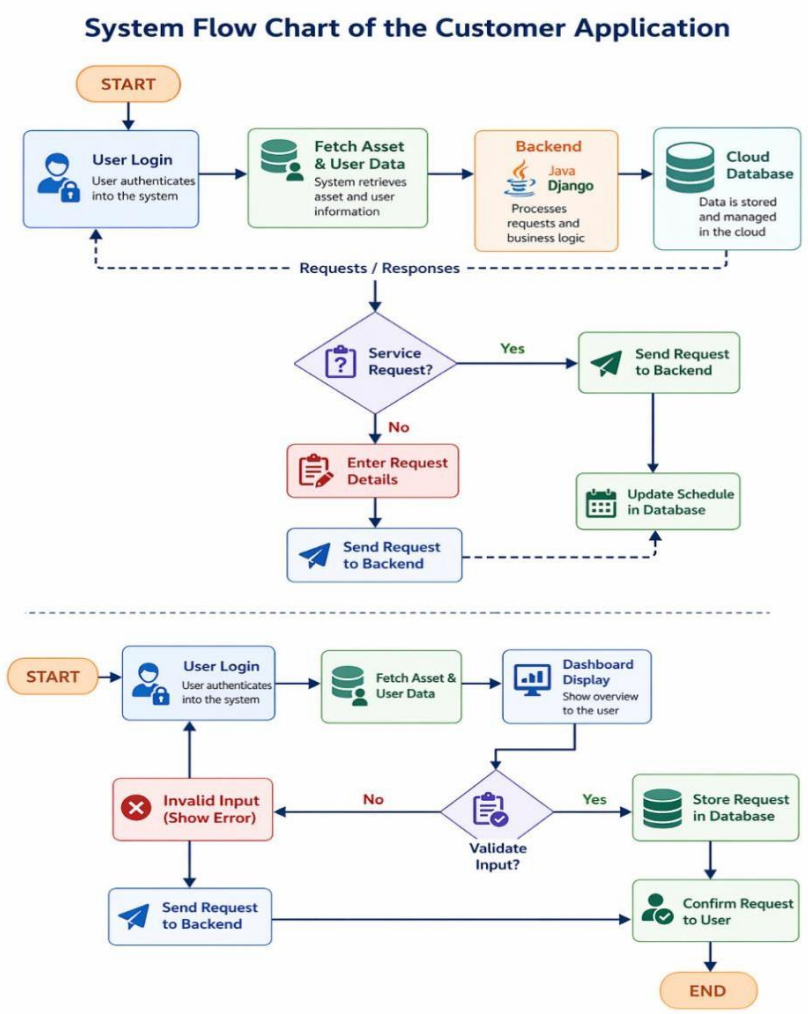

Figure 5.2: System Flow Chart of the Customer Application

Then, the flow continues to user actions, which provide users the ability to carry out actions like filing of service requests or rescheduling maintenance. For instance, when there is a service request filed, the first thing that the system does is validate whether the information entered by the user is valid or not. If there is any wrong input made by the user, an error message will be displayed to notify the user about the error. Otherwise, the information will be transferred to the backend for processing, and the information will be stored in the database.

When there is a successful processing of the information on the backend side of things, the next thing is updating the information on the database. In turn, this will lead to the generation of a confirmation message back to the frontend user. This whole flow process as illustrated in Figure

5.2 is important since it shows the interaction between the frontend, backend, and database.

Overall, the system flow chart provides a clear representation of how the application operates, from user interaction to backend processing and data storage. It ensures that all operations follow a defined sequence, improving efficiency, reducing errors, and enhancing the overall user experience of the application.

5.4 Choosing devices

Conforming to industry norms is crucial while designing a software application that is dependable and scalable. Throughout the process of development of the customer application at Axxonet Solutions, several industry standards have been adhered to. Industry standards facilitate error reduction and improve the overall development process.

Among the industry norms that have been adhered to is the principle of designing APIs that communicate between the front and back-end of the application according to the REST architecture. All the APIs have been designed using a consistent structure that includes the usage of standard HTTP verbs like GET, POST, PUT, and DELETE. RESTful API design ensures easy understanding, testing, and scalability. Besides, the documentation of the APIs has been done and standard responses have been set.

Furthermore, apart from desktop computers, the other types of devices that have been considered in order to improve accessibility include mobile devices, namely, smartphones and tablets. As users might want to access their applications while being on the move, the system has been designed with a user interface that allows for adjustments in case users change their devices. This makes it possible for users to be able to view asset information and make maintenance and service requests easily.

As another consideration for device choice, it has been decided to choose all devices that can provide users with an Internet connection. It is essential because users' requests for information will depend greatly on real-time interaction between the frontend application and the backend

servers hosted at the Avia Enterprises' headquarters.

In summary, the device selection has largely been driven by considerations for convenience. With the help of the selected devices, users will be able to access information in a flexible manner without being required to purchase any additional equipment.

5.5 System Unit Design

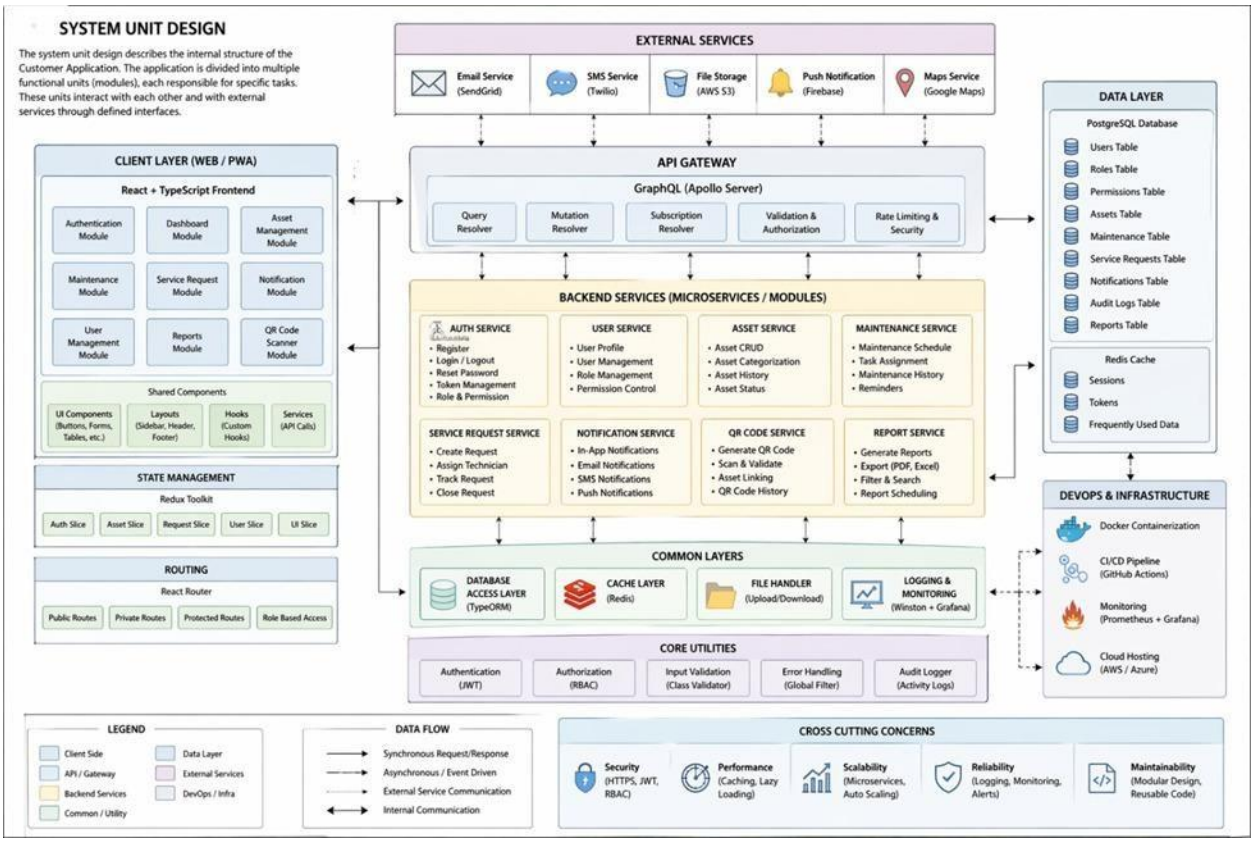

Figure 5.3: System Unit Design

Designing system units is critical in creating modules that are scalable and easy to maintain to ensure the successful completion of the customer application. As seen in the development of the application at Axxonet Solutions, the system had been broken down into various smaller units, where each is responsible for performing certain tasks. Through this, developers are able to work independently on the different units without affecting others.

Firstly, there is the Authentication Unit that provides services relating to authentication of the user logging into the system and validating them to have access. This ensures that only authorized individuals are able to use the system and have access to the relevant data they need. Among other functions, this unit ensures data privacy for the different users.

Secondly, the Asset Management Unit serves the purpose of showing details of the different assets of the lift to the customer. Among the details include unique identifiers of the asset and its status among others. This unit pulls data from the backend of the database and displays it to the customers.

The Maintenance Management Unit is designed to help deal with the AMC (Annual Maintenance Contract) information, including the details of planned and carried out service activities. Customers will be able to view when the next service is expected to be completed and what has already been done during the previous service periods. In such a way, customers will be prepared and will know their service schedules.

The Service Request Unit is used to send service request tokens to service providers in case something goes wrong with the lift. Customers will be able not only to make complaints, but also to track and monitor the status of requests and get updates from time to time. Moreover, rescheduling of services is possible within this module.

Finally, the User Interface Unit created with modern web technologies will make an interaction between user and the application more convenient. It includes all other units and shows users all necessary information about elevators and service activity. The design of these modules is modular and makes the system much more effective, easily maintainable, and scalable.

5.6 Standards

The implementation of the client side of the system is carried out according to certain standards, which help create reliable and easy-to-manage software solutions. Following industry standards

is important when building scalable applications that will be understandable and extensible in the future. During the internship, Axxonet Solutions adheres to standard practices at each stage, whether it involves programming, design, communications, or deployment.

One of the important standards for the current project concerns coding and its proper structure. In particular, the backend implemented in Django follows the Model View Template architecture. Appropriate conventions of naming and indentation, as well as coding in modules, are followed to make the code readable and understandable. Also, version control tools like Git are applied during coding.

Furthermore, the application uses certain standards regarding communications between the frontend and backend of the software. For example, it follows all API standards, namely the RESTful API standards, which make sure that any data exchange within the application takes place efficiently and consistently.

User interface and design standards are also essential aspects of the system. First, they are related to consistency and accessibility issues because all the elements of the application should be uniform and easy to operate. Moreover, using responsive design strategies will ensure that the app can be accessed via different devices without any difficulties.

Security standards should also be considered in order to enhance security. For instance, such standards include certain procedures concerning authentication and authorization. It can help to avoid problems associated with data misuse. In addition, secure data handling, input validation, and proper session management are also among these standards.

To summarize, it is crucial to pay attention to all kinds of standards discussed above in order to improve the system. In such a way, it can become reliable enough to operate in the real world environment belonging to Avia Enterprises.

5.7 IoTWF Reference Model Layers Mapping

Using the IoT World Forum (IoTWF) reference model, one gets an opportunity to better understand how data travels from physical objects to user applications. In terms of this application created for Axxonet Solutions, it will be quite possible to map the customer application to the layers described above in order to demonstrate how data on lifts is collected and delivered to the end-users. It will give insight into the system architecture and will show what role each component plays.

On the lower level, there is the device and connectivity layer, where lift systems, sensors, and other devices provide data on inspections, breakdowns, maintenance and send them to the central platform using network infrastructure. Even though the customer application does not interact with hardware directly, this layer is important because it serves as a source for collecting data that the customer application uses further for analysis.

Data accumulation and abstraction layers serve as storage places and process the collected data. In terms of this scenario, different data provided by various lift systems is accumulated in the databases of the enterprise and exposed to application through APIs. As for the use of APIs, it allows for

The customer application can be found at the higher levels of the stack in the application layer and the business layer. This includes the customer application software that offers functionality like asset tracking, maintenance, and service requests. Using this processed information, the business layer is able to derive additional insights like predictive maintenance and performance analytics. Therefore, it can be seen that the customer application runs at the application layer and takes advantage of all the other layers underneath.

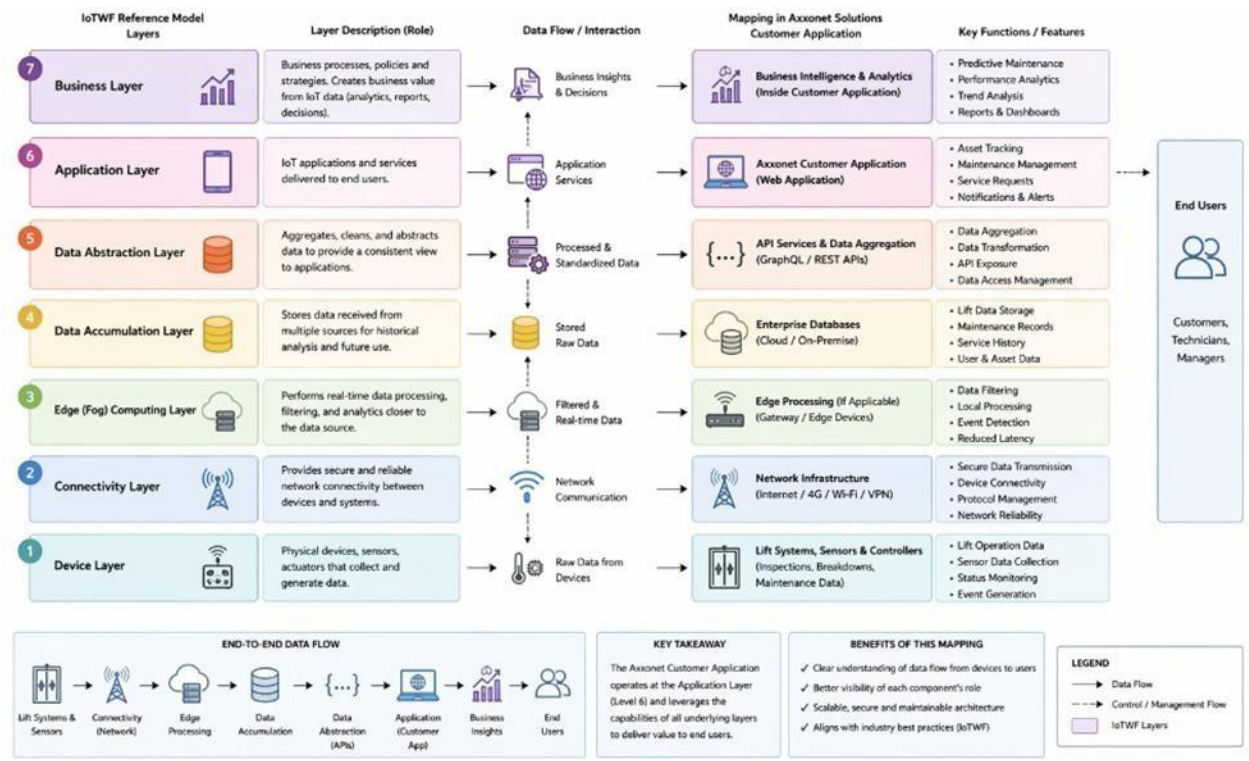

Figure 5.4: IoTWF Reference Model Layers Mapping

5.9 Communication Model

A communication model is important for defining the mode and the nature of the exchange of information among various components of the customer application. This is necessary to ensure a seamless interaction among the user interface, the backend services, and the database. The communication model adopted by the project is the client-server communication model, wherein the front end is the client while the backend processes requests and generates responses.

Communication takes place between the various elements of the client-server system. In order to perform an action like looking at the details of assets or creating a request for any type of service, the user makes a request from the client side. The backend process the request made from the frontend using APIs that adhere to RESTful architecture principles. In REST architecture, standard HTTP protocols like GET, POST, PUT, and DELETE are used. Once the backend processes the request and performs any other actions like database interactions, if required, a response is generated in structured format like JSON, which is easy to interpret by the client

system.

Besides the request-response paradigm, the system guarantees the security and integrity of information transmission. Authentication procedures are employed to ensure that users can be identified before they gain access to resources within the system. Validation measures have been put in place to handle input errors and avoid any form of interference within the system. The communication paradigm is scalable, and hence several users can access the system concurrently without any problems.

In conclusion, the communication paradigm ensures that all the different parts of the system coordinate seamlessly. The communication model offers a systematic path for information exchange, making it possible for real-time interaction within the system.

5.10 Level of IoT Implementation

The IoT implementation level of the proposed system refers to the way the application is designed concerning IoT technology usage in terms of connectivity, data processing, and interactions between devices and users. For this project, the IoT architecture used by the system belongs to the cloud-based deployment pattern, as the data collected by the physical devices such as elevators and sensors is transferred to a centralized system for processing and further analysis. The customer application is performed at the application layer based on processed data.

At the lower level, IoT-enabled devices, including lift sensors and monitoring systems, are responsible for collecting data about asset performance, condition, and maintenance. This information is transferred through the network to the cloud platform, where it is saved and processed by the system. Due to the centralized structure of the deployment, the data does not need to be analyzed locally and may originate from various sites. This increases the scalability of the application and its ability to handle many devices and users.

The middle layer involves data processing and management wherein data is converted into meaningful information. It entails the processes involved in filtering, validating, and analyzing

the incoming data in order to produce output data like maintenance plans and service alerts. APIs are employed in making the processed data available to the application layer thus facilitating seamless interaction between the backend services and the application layer.

The topmost layer involves the provision of customer application through which users get access to the system. With this, users will be able to see asset data, their maintenance plans, check their service timeline, and also create service requests. It enables users to make use of the IoT infrastructure in a more friendly manner.

Generally, it is an IoT implementation model that falls under the fourth level (Device -> Cloud -> Application), wherein data from devices is sent to the cloud, and applications utilize the data to offer certain services. The fourth level of implementation is ideal for practical use in industries because of its scalability and centralized control over data management.

5.11 Functional view

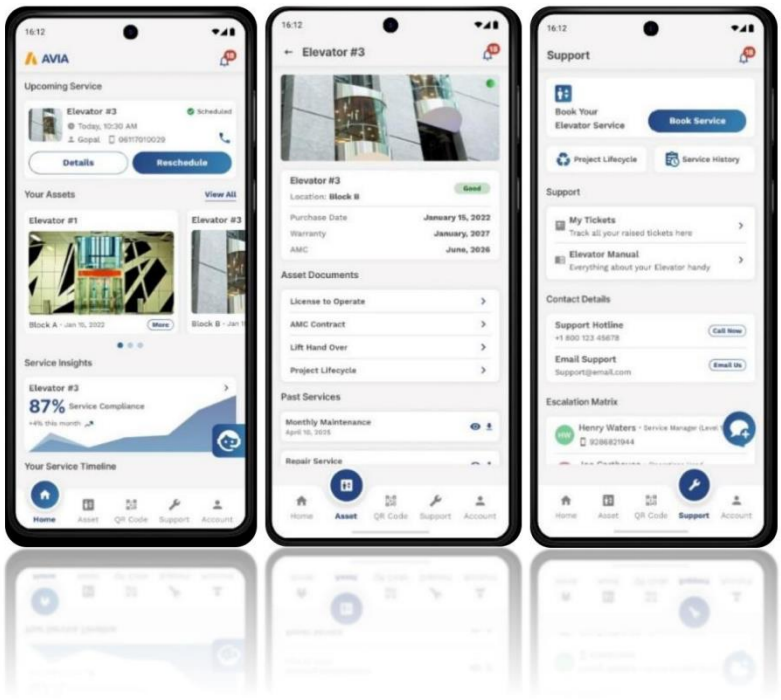

Figure 5.5: Functional view of Customer App

The functional approach towards the customer application is a description of how the system functions from the point of view of the user. Moreover, the system is arranged in terms of organization of its features that allow for an easy and effective usage. First of all, the user will be guided through the process from the beginning when logging into the system to further accessing services concerning lift management. The purpose of such an arrangement is making sure that all necessary procedures performed by the user on the platform created during the internship at Axxonet Solutions will be done without difficulty.

Once the user launches the application, he/she is provided with the login screen where the login details will be entered. They consist of username and password. After that, authentication of those credentials takes place in the backend. If the user's authentication succeeds, then he/she will be given access to the system.

Once the user is logged in, there are five primary tabs on the application, which serve different purposes: Home, Asset, QR Code, Support, and Account. The Home tab serves as the dashboard and presents an overview of information. Its components include Upcoming Services that help view services that are scheduled for the lifts. In addition, there is the component named Your Assets where users get a summary of their lifts. Another component under this tab is Service Insights, presenting analytical information. Also, there is the Service Timeline component that shows services from the past, current, and future. Other components under the Home tab include Open Support Tickets, listing currently opened tickets, and Need Help component for contacting customer support easily.

The Asset tab presents the assets owned by the user, which makes it possible to see information regarding each asset individually. The QR Code tab helps scan the QR code provided to each lift.

The Support tab helps manage service requests, and in this regard, users can check their tickets sorted according to their priority. As a result, users will be able to track their requests and ensure transparency in the process of handling services. Lastly, in the Account tab, users will be able to manage account settings by logging off from the application and selecting their preferred

language.

In general, this analysis of the functional aspects of the application shows how effectively several features can be integrated into one coherent system. Users will be able to get necessary information and perform necessary actions using the application.

5.12 Mapping of IoT Deployment Levels Using Functional View

Mapping of IoT deployment levels using the functional view of the customer application will help understand how different functionalities of the system match the various IoT deployment layers. Even though the application is mainly a web application, it works in an environment in which there are many devices generating data, which then is processed and displayed to end-users. With the alignment of functionalities of the system and IoT deployment layers, it becomes possible to understand the system better from the perspective of data flow, processing, and users' interaction in the environment provided by Axxonet Solutions.

Data generation happens at the device layer, where physical devices like lifts and sensors installed in them generate data about usage, performance, and maintenance of the devices. These data are transmitted to the connectivity layer through networking communication systems that transfer the data to backend servers. While the application itself may not interface directly with physical devices, it needs the data generated in the environment for proper functioning.

These levels, which include data processing and accumulation, are supported by backend software developed with the help of technologies such as Django. They collect, store, and process data transmitted by the devices. All the functional features, which include service insights, service timelines, and maintenance schedules, depend on such processing of data. Data can be organized in a manner accessible via APIs so that they can be retrieved by the frontend software.

As for the application layer, it can be considered a reflection of the functional layer since this is what determines the interaction of users with features provided. Such features as Home, Asset,

QR Code, Support, and Account are some of those that make up the application layer of our system. For instance, using the QR Code function, users can interact with the device and application layers at once as it enables them to scan the code and get instant access to the data about the asset.

Lastly, the business level revolves around gaining insights and increasing efficiency. The aspects like insights, asset tracking, and ticketing prioritization enhance decision making and service management. From the alignment of the functional perspective to the IoT implementation levels, the system shows how the flow of information from the hardware to the application layer is managed within the Avia Enterprises setting.

5.13 Operational Perspective

The operational perspective of the customer application entails the operational process in which the system operates in its deployment environment. It involves a discussion of how the application interacts with the users and its back-end services in a normal working environment. The operational perspective of the system involved ensuring efficiency and reliability to make sure that the end-user can easily use the application to coordinate lift maintenance.

The operational perspective of the application starts from the point when a user gains access to the application using a browser or an available device. The application then connects to its back-end services located at Avia Enterprises whereby data processing and storage occur. Once the user has logged in, the system will retrieve necessary information from the back-end in real-time. The purpose is to provide up-to-date data to the user when using the system.

The next phase of the workflow implies constant communication between the frontend and backend using API services. Once a certain action has been initiated by the user, the related request will be transferred to the backend where the backend will process the request, modify the relevant data if necessary, and transfer it back to the frontend. In addition, the operational model suggests the use of a number of specific technologies which facilitate and make all the processes more efficient.

Monitoring and troubleshooting represent another essential part of the operational model. The proposed app is expected to work properly in terms of exception handling. It will be able to deal with different kinds of exceptions, such as API errors, network problems, etc. Special tools have been designed to validate the input provided by users to provide feedback if there are some issues.

Chapter 6

Hardware, Software and Simulation

In this chapter, I will introduce the hardware, software, and simulation used in developing the customer application. Even though this particular application focuses on software, there is an ecosystem beyond just software that contains hardware, such as lift systems and monitoring devices. It is necessary to pay attention to both hardware and software factors to ensure the development of a scalable and sustainable system within the context of PaaS services offered by Avia Enterprises.

Firstly, there is no need for any specific hardware device in this case because standard computers were enough to conduct programming and testing and execute the application. Moreover, in practical life, the application does not communicate with hardware elements, which include lift systems and monitoring devices directly. The application has no direct control of hardware systems; control functions are performed at the infrastructure level.

Software development is one of the essential parts of the project development process. It consists of the frontend that is built with the help of React technology and Material UI framework, which will provide the application with responsive design. The backend part is developed using Django, which will help to implement business logic and build up API that can be used to facilitate communication between the front and back end.

Simulation of the system means testing its performance in the development environment before implementation. There are several types of tests that need to be performed for testing use cases. Such use cases include user authentication, getting access to certain assets, generating requests for services, API interactions, etc. Developers use mock data to test each aspect separately and fix possible errors and problems.

6.1 Hardware Requirements

The hardware requirements of the customer application are relatively low, since the product consists mainly of software and is aimed at functioning on common hardware. The development and testing of the application is performed with the help of regular desktop and laptop computers possessing sufficient computing resources, memory capacity, and space for supporting development tools, server infrastructure, and web browsing applications. Such hardware provides convenient conditions for developing the application effectively.

As for end users, the system does not demand any specific hardware as well. Customers may use any type of computer (desktop, laptop, tablet, smartphone), provided that the latter runs some web browser and possesses connection to the Internet.

With respect to the operational environment, the software will operate indirectly with hardware elements like lifts and other systems. Such hardware could consist of sensors and other controlling devices used to assess the efficiency of lifts as well as any maintenance needs. The software does not actually run the hardware, but it uses the information collected from those systems.

Another critical component in the environment of this system would be hardware that ensures effective communication between all system components. Internet access, routers, and the cloud would be vital to ensure instant data exchange between the frontend, backend, and databases. The hardware requirements would be straightforward and affordable.

6.2 Software Development Tools

In order to develop the customer application, a number of different software tools have been used. The first tool that will be considered is called Frappe. This tool acts as a basis for developing enterprise resource planning (ERP), which allows organizing all necessary information about business transactions. One of the main advantages of using Frappe is the ability to organize

backend architecture of applications as well as provide functions like database management and API generation.

Frontend development requires new technologies like React and Material UI (MUI). React allows creating components that will help develop applications using a component-based approach. As for Material UI, it will ensure that a uniform style and design will be preserved throughout the entire app and will contribute to providing a quality user experience through the creation of a responsive and user-friendly interface.

The backend development and API processing is enabled with the help of technologies and frameworks that are already implemented into Frappe software. APIs can serve as channels for communication between the front and back of the application as well as provide efficient data transfer. Also, in addition to the tools implemented in Frappe, there are technologies, such as Django, which may also be helpful when developing a website.

6.3 Simulation

Simulation helps verify the working of the customer application before it gets launched in a practical setting. The simulation helps validate the application based on how the application would react with controlled data and simulated scenarios in order to check whether all aspects of the system run smoothly without causing any effect on live data and activities.

The following are some of the simulated use cases which need to be tested: login authentication, asset information, maintenance schedules, service requests, and re-scheduling of services among others. Through such simulations, possible errors in data management, inconsistencies in the interface, or problems with the APIs can be detected and sorted out to make the application reliable.

Another critical area is API simulation wherein request and response between the frontend and the backend are simulated. Mock data are frequently used in simulating responses of the backend especially in the early stages of the application where live APIs would not yet be available. This

allows for an independent development of frontend through React while ensuring that the frontend works correctly with the backend services.

Apart from functional tests, there is also a simulation of the application's performance and error handling capabilities. The performance of the application is tested under various circumstances which include but are not limited to many user requests, poor internet connection, and invalid inputs to name a few. Validation and error message are employed to test how the application reacts to such situations.

The simulation process guarantees that the application will be stable and efficient after being deployed.

Chapter 7
Evaluation and Results

There are a number of factors that were used to analyze the customer application including its performance, function, and usability. Generally, the system performed well in terms of providing correct information regarding assets, scheduling maintenance of assets, and managing service requests. It is important to note that the application did well by allowing users to log in, view their assets, maintain history through service tickets, and also make service requests effortlessly. Seamless interaction between the front-end and back-end was witnessed since information was processed and presented seamlessly.

Performance-wise, there were a few areas where the application did not do well in some instances. There were delays in responding to input commands and validating entries. Nevertheless, in relation to usability, the application performed quite well since it provided a good user interface where users could navigate easily and access all relevant information.

7.1 Test points

The test points form the checklist for the functionalities of the system that need to be verified so as to determine the effectiveness of the application. In this regard, test points assist in determining the key areas of testing in the application, which cover the various functionalities and non-functional aspects. The customer application will undergo testing at several points to guarantee its effectiveness.

One of the key modules in testing will involve user authentication, which is important in terms of the overall performance of the application in relation to security issues and user permissions. In terms of user authentication, some of the test points will include verifying that a user logs into

the system correctly; entering incorrect usernames or passwords; and verifying login and logout features such as login persistence and auto log out on inactivity. Other tests will involve verifying validation of the inputs, including null fields.

Asset management is another crucial module, where it should be ensured that the assets of the logged-in user are appropriately displayed in the system. The test points for this module would be validation for displaying relevant assets for the user, proper display of asset information including their ID, location, and specifications, along with consistency checks across different pages. Edge cases such as users having no assets as well as those with multiple assets would also be taken into consideration.

Maintenance and service management would be another key module. The testing point here would be the validation for proper scheduling of maintenance and services based on AMC. For instance, the system should be able to properly identify upcoming, ongoing, and completed services, as well as the ability to schedule new services and reflect updated schedule changes.

Ticket generation is a vital component which enables users to file their complaints. Testing scenarios include confirming that the token generation process works effectively, filling up all necessary fields, confirming that the token has been submitted successfully, and ensuring that the token is found within the support portal. Other test scenarios include testing for invalid data entry, duplicate tokens, and confirmation of token statuses.

QR code is a unique capability of the application. Testing scenarios include scanning the QR code, confirming that the scanned code refers to the relevant asset, and enabling users to submit their service requests through the scanned asset. The other testing scenarios involve scanning invalid QR codes.

Apart from the functional test points, non-functional test points are also taken into account. This would involve the examination of how responsive the user interface is in terms of performance when different devices are used, analysing the performance of the system with regard to multiple user requests, and verifying the error-handling process.

Through all these processes, the software can be assured to be fully functional, reliable, and friendly.

7.2 Test plan

The test plan describes the detailed process followed in the validation of the functionality, performance, and reliability of the application intended for use by customers. The test plan is essentially a roadmap that shows how the testing process is done, which features are tested, and the basis on which they are evaluated. Test plans are important because they enable the tester to ensure that all parts of the software being tested are adequately examined to identify any weaknesses that may exist.

The primary goal of the test plan is to ensure that all functions intended in the software are performed as expected, including user authentication, management of assets and maintenance, management of service requests, and operations using QR code. The scope of testing includes all critical modules in the application, such as the login module, dashboard, details of the assets involved, timeline of services, and support tickets. Notably, some aspects are excluded from the scope of testing, particularly advanced security testing and benchmarking of performance levels.

The proposed approach to testing involves the use of different tests to provide thorough validation. It is necessary to conduct functional tests to confirm the proper functioning of each component, regression tests to prove that an update does not impact the operation of features implemented previously, and smoke tests to establish the stability of the system in general. The method of testing will mainly include manual testing and API testing of all modules.

The setup of the testing environment will involve providing the conditions similar to those that would be faced by users. The application will be checked in the QA or staging environment with the help of various browsers, e.g., Chrome, Firefox, and Edge. Moreover, different operating systems will be taken into account to assess the cross-platform compatibility of the system. The right test data, which includes valid and invalid user credentials, assets, AMC data, and service requests, will be created.

A number of tools such as test management tool, API testing tool, and debugging tool are included in the testing support area. The roles and responsibilities of team members for performing testing are well-defined, and there are specific people to conduct testing related tasks like test case designing, executing tests, fixing defects. The entry criteria for testing involves having an executable, finalized set of requirements and an available test environment. The exit criteria include completion of test case executions, resolving all important defects, and readiness of the system to go live.

Moreover, the schedule involved in the testing process has also been determined, which indicates the duration for conducting different testing activities like testing planning, test executions, bug fixes, etc. Deliverables such as test cases, test execution reports, defect reports have been prepared and documented in the testing process.

7.3 Test result

The test results were collected after conducting a series of tests on the functionality of the login module of the customer application. In total, 12 test cases were generated and conducted. These test cases included different scenarios such as a successful login, unsuccessful login due to incorrect login credentials, validation of user interfaces, testing of password recovery capabilities, and other aspects related to input validation.

The results of the test case execution demonstrate that all the test cases have been successfully completed. The login module works properly for all possible scenarios. Valid login credentials result in successful logins and redirect users to the homepage. For example, an incorrect password leads to a message about incorrect login credentials. Similarly, an unregistered email and a null input in any of the fields cause a message to appear stating that the entered value is incorrect.

Other additional test cases were based on the usability of the website, which included validation of various UI components such as form input fields, button objects, hyperlinks, and application logo. Navigation operations like “Forgot Password” and navigation back to the login page

have been successfully tested. The password reset feature was also verified, and users are receiving confirmation upon entering a correct email address.

Thus, it is evident that the website can handle various situations both functionally and through validation processes without any difficulties or errors. No test failures occurred during the whole testing process, which means the login module is implemented successfully. Test case table with all details about test cases and their results is presented below.

7.4 Insights

Among others, the testing process resulted in some useful observations about the way the customer application behaved and performed. One of them is that a carefully thought out validation scheme makes the application much more reliable. Indeed, the application demonstrated a robust approach to handling input in various cases – from correct and incorrect input to edge cases where special measures are required. All these cases are treated in a way that ensures users get proper feedback regardless of the input.

Another observation has to do with the importance of user interface design and its impact on the quality of interaction between user and software. Testing revealed that clear and unambiguous error messages, well-aligned user interface items and intuitive navigation make an application very convenient to work with. Thus, the login screen, password reset functionality and navigating between different screens did not create any issues for testers to use the application.

The testing phase also stressed the importance of conducting thorough tests. Through the testing of numerous situations, such as edge cases like inputting empty values and entering incorrect credentials, the system could be tested comprehensively. This ensured that all problems were captured and rectified even if they were small. Moreover, the testing phase proved the importance of testing techniques, which include developing test points, creating test plans, and analyzing the outcomes.

On the other hand, the findings revealed the importance of integration in the system. The

coordination between the front-end and back-end of the system allowed for the proper processing of information and display. In case there was any delay in the response to API requests, it was quickly detected through testing and optimized accordingly.

These factors all combine to enhance the quality of the application as a whole. They allow for a deeper understanding of how the systems work together and can be improved upon in the future.

Chapter 8

Social, Legal, Ethical, Sustainability and Safety Aspects

In the process of creation of modern software products, it is necessary not only to pay attention to technical characteristics but also to take into account the consequences for society, the legal issues, the ethics of the product, sustainability of use, safety, etc. The customer lift management application created in the course of the internship at Axxonet Solutions is designed to work in such an environment where all these aspects play a significant role. Taking into account these aspects allows you to build an effective product.

Socially, the application positively influences relations between users and managers, providing additional quality and comfort for customers. From a legal point of view, it should also comply with many requirements, such as data protection rules, software license usage, etc. Ethics plays a huge role here as well.

On the subject of sustainability, the program is geared towards maximizing resource efficiency by facilitating digital operations and decreasing the need for paperwork. Utilizing cloud applications like Avia Enterprises also plays an important role in limiting environmental damage. Safety is yet another crucial aspect, particularly when it comes to elevators. The program guarantees safety and security due to its efficient handling of data and maintenance records..

8.1 Social Issues

The creation of the customer application for lift management carries substantial social implications in terms of the improved interaction of the service provider and client. Previously, the clients did not possess much information concerning their lift system, its performance, and servicing. The introduction of a digital platform makes the process of interaction transparent and allows the user to quickly gain access to the data about the lift system and services. It increases transparency and eliminates the need for a customer to engage in any manual actions.

The developed system is also inclusive since it provides users of diverse technical skill level with a convenient interface. Inclusiveness is crucial due to the fact that users may lack experience using similar platforms and might encounter difficulties when operating the system. It allows the application to be used within both residential and commercial premises where users might be unfamiliar with the system. It facilitates a transition to more advanced technologies in society.

Another important social benefit would be an increase in the efficiency of the service process. As the customer will be able to see the schedule for maintenance, re-schedule the servicing of the lift, as well as submit requests for services, delays will be reduced. This will improve the quality of services provided by the company. The company will have better control over the maintenance process, and the possibility of system failures will decrease significantly.

The application will also allow improving the level of accountability among individuals working on the project. As all requests and maintenance processes will be documented, this will make the monitoring and checking of the work of people working with the lifts much easier. This will ensure the best practices, increasing the level of satisfaction of customers.

Finally, the application will promote the digitalization of our society, which is an extremely important contribution that can be made with the help of the project. Lift management is an industry that was always rather traditional, but now it needs to evolve and become part of the digital environment.

8.2 Legal Considerations

Legality is one of the main factors that must be considered during both the creation and implementation processes of the customer application. In the first place, it is necessary to make sure that all data will be stored and transferred safely. Since the application will be storing personal information about its customers, assets, and services, data protection law and regulations must be followed.

It is also crucial to consider legality from the point of view of licensing and software usage. In this case, it must be mentioned that the application uses several technologies, such as React, among others, which are also open source. Therefore, it is crucial to follow the license terms in order not to have any legal problems.

Apart from complying with the legal guidelines, the application has to be in accordance with the contractual obligations of service providers and their customers. As a reminder of the maintenance schedule and services to be performed according to the agreements made within the framework of AMC, it is extremely important that all the information shown by the system be precise and accurate. Otherwise, there will be possible legal issues or complaints from customers.

Moreover, in order to comply with legal requirements, the system provided by Avia Enterprises needs to meet certain standards of the organization or the industry. It will be the obligation of the organizations implementing the application to follow those standards, which means keeping adequate records and providing accurate data in accordance with the standards of software development.

Finally, it is necessary to make sure that there is legal responsibility for any problems or mistakes that may occur while using the application.

8.3 Ethical Issues

There are a number of ethical issues that need to be considered in software application development, especially for applications dealing with personal information and service provision. The main issue related to ethics is ensuring privacy and confidentiality of all user-related data. The software needs to be developed in such a way as to ensure the security of the data and prevent any kind of misuse. Moreover, developers should not collect any unnecessary data from users.

Another important ethical issue relates to transparency. The software gives users the ability to look at various types of information relating to the assets they have. They get to see when the maintenance is due and what services have been provided in the past. Providing users with this type of information in a timely and precise manner is important.

Accessibility and fairness are equally essential. The application itself needs to be inclusive and accessible to everyone, despite the level of knowledge or experience in information technology that each user has. It must be developed in a way that will allow any individual who uses it access to the same benefits without being discriminated against.

The next ethical issue to consider is related to the proper use of technology. The application should not be used in such a way that it would benefit only one party at the expense of others. Such things as tracking service requests, which was mentioned above, should not be used to exploit anyone.

Finally, the ethical responsibility also includes professional behavior while working on the project. Being a part of Axxonet Solutions, one needs to be able to work professionally and efficiently within the organization and with other people involved in the project.

8.4 Aspects Related to Sustainability

Sustainability is one of the key aspects considered during software development nowadays. The customer application promotes sustainability because it is designed to digitize various activities that were previously conducted manually, including documentation and services monitoring. Thus, using the application decreases resource consumption and waste production.

Additionally, the application allows for efficient maintenance management and effective use of available resources. With the help of the application, it becomes possible to schedule appropriate maintenance times in advance and provide service when necessary, ensuring optimal resource utilization and decreased energy consumption during the provision of maintenance services.

In addition to software applications, cloud-based solutions contribute to sustainability. For instance, the platform of Avia Enterprises enables users to conduct all necessary activities without wasting additional energy on running a local infrastructure. Thus, using such a platform contributes to sustainability significantly.

Additionally, the application promotes sustainability by enhancing the longevity and efficiency of elevator systems. The proper maintenance and servicing of these systems through the use of the application ensure that there will be fewer breakdowns and that the equipment will last longer, leading to less waste creation.

Lastly, sustainability in software development is also achieved through scalability and maintainability. With the use of technologies like React, the application has been developed in such a way that it is efficient and can easily adapt to future changes.

8.5 Safety Features

Security is one of the most crucial aspects of developing any application dealing with lift management systems. It affects the overall well-being of the users who may be at stake due to some technical failure. The safety of users increases with this application due to timely provision of necessary information regarding the maintenance services required.

Raising the number of service request tokens is yet another security aspect associated with lift management software. With the help of this, customers will be able to send their service requests without any delay, thus preventing any further damage to the system.

Ensuring data safety is a significant task when developing applications for lift management. Since the system stores customer information, it becomes very important to ensure its safety by employing various encryption methods.

Further, the application needs to be able to deal with any sort of errors and problems that might arise. Validation and error messages as well as other safety measures make sure that there will be no sudden shutdowns and that everything works smoothly. This is especially important in cases where there might be serious outcomes because of a failure.

Lastly, safety includes the entire process of developing an application. Applying best practices, thorough testing, and organization rules and procedures will help make sure that the application is safe and reliable. In this way, all the necessary factors concerning safety are being taken into account.

Chapter 9

CONCLUSION

The six-month internship offered an extensive opportunity to get practical experience in the area of full-stack development by creating a customer app for lifts management in the platform-as-a-service architecture of Avia Enterprises. The process of learning included stages that started from basics and proceeded towards the implementation of a specific project. As a result, the theoretical and practical knowledge of modern web development was gained.

The first stage involved the development of a comprehensive knowledge base, including programming and web technologies such as HTML, CSS, JavaScript, React, Java, Python, and Django. The knowledge obtained at this stage proved to be very important as it made it possible to understand the principles of frontend and backend web development.

The second stage was mainly devoted to the development of personal projects like websites. During this period, there was an opportunity to improve skills related to UI/UX design, solving issues, and developing responsive designs.

The major aim of the internship was to create an application that would serve customers regarding lift management. The application had some features that included the ability to check asset details, AMC based maintenance planning, scheduling and rescheduling of services as well as requesting service (tokens).

As far as the technology applied in creating the application is concerned, React framework combined with MUI (Material UI) was used to make a responsive, adaptable application user interface possible. The application was created based on the concept of components, meaning that each component could be modified individually without affecting other parts. Moreover, hooks like useState and useEffect facilitated effective API integration.

Development of the project took place within four stages. These comprised requirement analysis,

system design, implementation and testing. Some difficulties were faced during development. Nevertheless, all problems could be easily solved due to debugging and further learning process.

Besides technical competencies, the internship also highlighted the significance of managing projects, assessing risks, and considering social, legal, ethical, sustainable, and safety aspects. It gave an all-around view of software engineering, where the systems are not only expected to perform well but should also adhere to the set criteria and be responsive to the expectations of society.

In conclusion, the internship played a vital role in enhancing the personal and professional skills of the intern. First, it helped to gain practical industry experience. Apart from learning technical skills, the internship also imparted complete knowledge about the whole software development process. As a result, the internship prepared the intern for a career in full-stack development.

In summary, this internship period with Axxonet Solutions was extremely beneficial in bridging the gap between theory and practice. This experience has taught me that the application of technology in real life is possible through the customer application on the Avia Enterprises platform. The skills and experience acquired during this period will be very helpful in my future endeavors.

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Base Paper:

The primary paper referenced from the References section:

[1] G. Brito and M. T. Valente, "REST vs GraphQL: A Controlled Experiment," Proc. IEEE Int. Conf. Software Architecture (ICSA), 2020, pp. 81 – 9

APPENDIX

1. Publications:

2. Project Report -Similarity Report

3. Images of the Project:

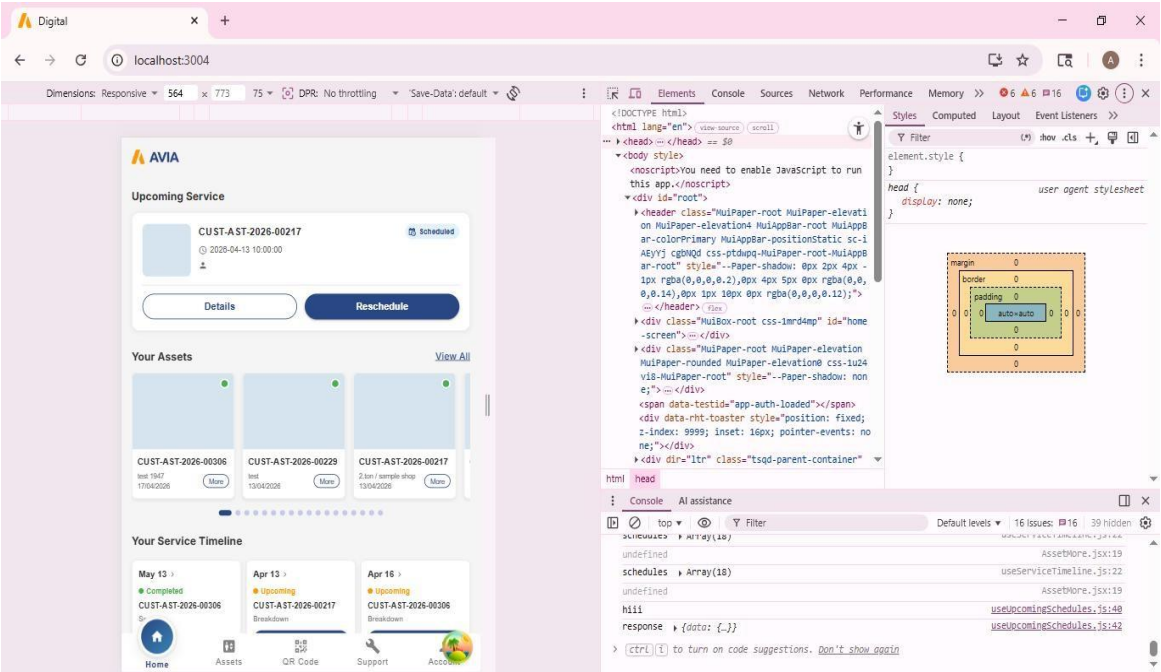

Fig A.1: Snippet of the Application